CSE 536 – Advanced Operating Systems

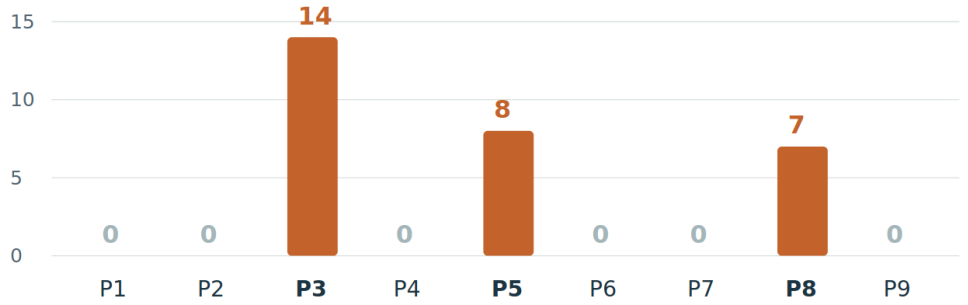
Fearless Conspiracy

Thread Privacy

Private chat reached only Mafia players, but readable server thread statistics showed who was receiving the messages.

Private chat left a visible pattern

Five Mafia-chat commands were sent in one test window. The bars show the largest increase in voluntary context switches across each player's connection threads.



The graph shows that the thread leaked which voluntary context switches were happening across each thread, meaning Mafia status could be inferred

What did I learn from this?

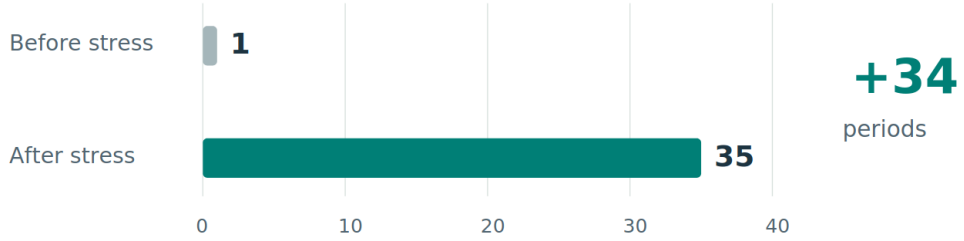
I learned that the private messages can leave clues in the thread activity similar to the original werewolves. I learned that to fix this, I could toggle /proc visibility using hidepid=2 which hid other users' process directory.

Cgroup resource limits

To stop players from potentially hogging resources to crash or stall the game, I gave each Linux account one combined CPI, memory, and task budget.

CPU stress activated the quota

The player had a CPU budget of 25% of one core. Linux made the player's tasks wait when that budget was used. The bars show the cumulative number of quota periods with throttling before and after the stress test.



The graph above shows that while running a CPU intensive program under one of the players accounts, it recorded 34 additional periods of throttling meaning that the CPU budget was enforced

What did I learn from this?

I learned that when you have multiple players that are sharing any kind of resource, you have to be wary and limit it in advance to give bad actors less freedom with their possible attack vectors.

The biggest lesson I learned was to control what the players can control and consume.